CS & IT

Algorithms

DPP: 1

Sorting

Q1 What is the recurrence relation of insertion sort when the array is almost sorted with P element?

(A) $T(P) = T\left(\frac{P}{2}\right) + O(1)$

(B) $T(P) = T\left(\frac{P}{2}\right) + p^2$

(C) $T(P) = 2T\left(\frac{P}{2}\right) + p$

(D) $T(P) = 2T\left(\frac{P}{2}\right) + 0(1)$

Q2 Consider the following array A with 8 elements:

A

70	60	20	50	40	5	19	21
0	1	2	3	4	5	6	7

What is the index value of elements 60 after 3rd pass of selection sort?

Q3 Consider the following array with 8 elements:

50	60	90	65	55	45	85	12

What is result after 3rd pass of bubble sort?

(A) 50, 60, 55, 45, 12, 65, 85, 90

(B) 12, 45, 50, 60, 90, 65, 55, 85

(C) 90, 85, 65, 50, 60, 55, 45, 12

(D) 50, 55, 45, 60, 12, 65, 85, 90

Q4

A

10	20	50	60	70	65	55	25	15

How many swaps are needed to sort the array by using is sertion sort _____?

Q5 Bubble sort is

(A) In place sorting technique

(B) Outplace sorting technique

(C) Unstable sorting technique

(D) Stable sorting technique

Q6 What is the time complxity of selection sort in best case, average case and worst case respectively is:

(A) $O(n), O(n\log n), O(n^2)$

(B) $O(n), O(n^2), O(n^2)$

(C) $O(n^2), O(n^2), O(n^2)$

(D) $O(n), O(n\log n), O(n\log n)$

Q7 How many swaps are needed in selection sort to sort n element in worst case?

(A) n –1

(B) $\frac{n(n-1)}{2}$

(C) n

(D) n^2

Q8 Consider the following elements:

101	56	934	555	8	12	785	23	5	999

What is the results after 3rd pass of padix sort?

(A) 56, 23, 12, 8, 5, 101, 55, 785, 934

(B) 56, 23, 12, 8, 5, 101, 555, 785, 999, 934

(C) 5, 8, 12, 23, 56 101, 785, 555, 934, 999

(D) 5, 8, 12, 23, 56, 101, 555, 785, 934, 999

Answer Key

Q1	(D)
Q2	6
Q3	(D)
Q4	16
Q5	(A, D)
Q6	(C)
Q7	(A)
Q8	(D)

Hints & Solutions

Q1 Text Solution:

$$T(P) = 2T\left(\frac{P}{2}\right) + O(1)$$

Master's theroem

T(P) = O(P)

Q2 Text Solution:

i/P = 70, 60, 20, 50, 40, 5, 19, 21

Take first min & swap it to first place

Pass-1: 5, 60, 20, 50,40, 70, 19, 21

Pass-2: 5, 19, 20, 50, 40, 70, 60, 21

Pass-3: 5, 19, 20, 50,40, 70, 60, 21

In dex value of element 60 is 6.

Q3 Text Solution:

55, 45, 85, 12

65, ~~90~~, ~~90~~, ~~90~~, ~~90~~, 90

I/P: 50, 60, 90, 65, 55, 45, 85, 12

45

55, ~~65~~, 65, 12, 85

Pass:-1 50, 60, ~~65~~, ~~55~~, ~~45~~, ~~85~~, ~~12~~, (90)

45

55, ~~60~~, 60, 12, 65

Pass:-2 50, ~~60~~, ~~55~~, ~~45~~, ~~65~~, ~~12~~, (85, 90)

Pass:-3 50, 55, 45, 60, 12, (65, 85, 90)

Q4 Text Solution:

- | 10 | 20 | 50 | 60 | 70 | 65 |
  0 0 0 0 1
- | 10 | 20 | 50 | 60 | 65 | 70 | 55 | |
  3
- | 10 | 20 | 50 | 55 | 60 | 65 | 70 | 25 |
  5
- | 10 | 20 | 25 | 50 | 55 | 60 | 65 | 70 | 15 |
  7
- | 10 | 15 | 20 | 25 | 50 | 55 | 60 | 65 | 70 |

Total Swaps = 1 + 3 + 5 + 7 = 16

Q5 Text Solution:

Bubble sort is inplace & stable sorting algorithms.

Q6 Text Solution:

Time complixity of selection sort is $O(n^2)$ in every case.

Q7 Text Solution:

Number of swaps in selection sort in every caseis (n-1). Selection sort is best for swaps.

Q8 Text Solution:

Pass-1: 101, 12, 23, 934, 555, 785, 5, 56, 8,999

Pass-2: 101, 5, 8, 12, 23, 934, 555, 56, 785, 999

Pass-3: 5, 8,12, 23, 56, 101, 555, 785, 939, 999